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1. Let $A = \{1, 2, \{3, 4\}, \{5\}\}$. Decide which of each of the following statements are true or false.
   (a) $1 \in A$
   (b) $\{1\} \in A$
   (c) $\{5\} \in A$
   (d) $\{1, 2\} \in A$
   (e) $\{1, 2\} \subseteq A$
   (f) $\emptyset \in A$

Solution: Enter your solution here:
(a) $1 \in A$ **True**. The element 1 is listed explicitly as a member of A.

(b) $\{1\} \in A$ **False**. The set $\{1\}$ is not one of the four elements listed in A; only the element 1 itself belongs to A.

(c) $\{5\} \in A$ **True**. The set $\{5\}$ is listed explicitly as a member of A.

(d) $\{1, 2\} \in A$ **False**. The set $\{1, 2\}$, taken as a single object, is not one of the four elements listed in A.

(e) $\{1, 2\} \subseteq A$ **True**. Both $1 \in A$ and $2 \in A$, so every element of $\{1, 2\}$ is an element of A.

(f) $\emptyset \in A$ **False**. The empty set is not one of the four elements listed in A.

2. Consider the following logical equivalence:

Theorem. For all logical statements P, Q, and R, we have

$$(P \Rightarrow Q) \Rightarrow R \equiv \big((\neg P) \Rightarrow R\big) \wedge (Q \Rightarrow R).$$

Prove Theorem 1 using previously established logical equivalences.

Solution: Enter your solution here:
Proof. We assume that P, Q, and R are logical statements, and we will show that $(P \Rightarrow Q) \Rightarrow R$ is logically equivalent to $\big((\neg P) \Rightarrow R\big) \wedge (Q \Rightarrow R)$ by simplifying the right-hand side with a chain of logical equivalences until it matches the left-hand side.

$$\begin{aligned}
(P \Rightarrow Q) \Rightarrow R &\equiv \big((\neg P) \Rightarrow R\big) \wedge (Q \Rightarrow R) && \text{Given} \\
&\equiv (\neg\neg P \vee R) \wedge (\neg Q \vee R) && \text{Implication (applied twice)} \\
&\equiv (P \vee R) \wedge (\neg Q \vee R) && \text{Double Negation} \\
&\equiv (P \wedge \neg Q) \vee R && \text{Distributive} \\
&\equiv (\neg\neg P \wedge \neg Q) \vee R && \text{Double Negation} \\
&\equiv \neg(\neg P \vee Q) \vee R && \text{De Morgan} \\
&\equiv \neg(P \Rightarrow Q) \vee R && \text{Implication} \\
&\equiv (P \Rightarrow Q) \Rightarrow R && \text{Implication}
\end{aligned}$$

Since each step above is a logical equivalence, we conclude that

$$(P \Rightarrow Q) \Rightarrow R \equiv \big((\neg P) \Rightarrow R\big) \wedge (Q \Rightarrow R).$$

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3. Use set builder notation to write down the following sets:
   (a) The set A of all odd positive integers.
   (b) The set B of all real numbers strictly larger than π.
   (c) The set C given by
   $$C = \left\{\ldots, 2^{-4}, 2^{-3}, 2^{-2}, \frac{1}{2}, 1\right\}.$$

Solution: Enter your solution here:
(a)

$$A = \{x \mid x = 2k + 1,\ k \in \mathbb{Z},\ k \geq 0\}$$

(b)

$$B = \{x \in \mathbb{R} \mid x > \pi\}$$

(c) Each element of C is a power of 2 with a nonpositive integer exponent, since

$$\ldots, 2^{-4}, 2^{-3}, 2^{-2}, 2^{-1} = \tfrac{1}{2}, 2^{0} = 1.$$

$$C = \{x \mid x = 2^{n},\ n \in \mathbb{Z},\ n \leq 0\}$$

4. Consider the following statements:
   (a) There exists a rational number that is less than or equal to -1010.
   (b) For all integers x and y, we have $xy > x$.
   (c) For all real numbers ϵ, there exists a real number δ such that if $|x - \pi| < \delta$, then $|\sin(x)| < \epsilon$.

   For each statement, complete the following three tasks:

   - Rewrite the statement in symbolic form.
   - Write its negation in symbolic form.
   - Write a useful negation of each of the following statements as English sentences, without symbols for quantifiers, but where you can use symbols as they were used in the original statements.

Solution: Enter your solution here:

(a) There exists a rational number that is less than or equal to -1010.

Symbolic form:

$$\exists x \in \mathbb{Q}\ (x \leq -1010)$$

Negation (symbolic form):

$$\neg\big[\exists x \in \mathbb{Q}\ (x \leq -1010)\big] \equiv \forall x \in \mathbb{Q}\ (x > -1010)$$

Useful negation (English sentence): Every rational number x satisfies $x > -1010$.

(b) For all integers x and y, we have $xy > x$.
Symbolic form:

$$\forall x \in \mathbb{Z}\ \forall y \in \mathbb{Z}\ (xy > x)$$

Negation (symbolic form):

$$\neg\big[\forall x \in \mathbb{Z}\ \forall y \in \mathbb{Z}\ (xy > x)\big] \equiv \exists x \in \mathbb{Z}\ \exists y \in \mathbb{Z}\ (xy \leq x)$$

Useful negation (English sentence): There exist integers x and y such that $xy \leq x$.

(c) For all real numbers ϵ, there exists a real number δ such that if $|x - \pi| < \delta$, then $|\sin(x)| < \epsilon$ (for every real number x).
Symbolic form:

$$\forall \epsilon \in \mathbb{R}\ \exists \delta \in \mathbb{R}\ \forall x \in \mathbb{R}\ \big(|x - \pi| < \delta \to |\sin(x)| < \epsilon\big)$$

Negation (symbolic form):

$$\neg\big[\forall \epsilon \in \mathbb{R}\ \exists \delta \in \mathbb{R}\ \forall x \in \mathbb{R}\ \big(|x - \pi| < \delta \to |\sin(x)| < \epsilon\big)\big] \equiv \exists \epsilon \in \mathbb{R}\ \forall \delta \in \mathbb{R}\ \exists x \in \mathbb{R}\ \big(|x - \pi| < \delta \wedge |\sin(x)| \geq \epsilon\big)$$

Useful negation (English sentence): There exists a real number ϵ such that for every real number δ, there exists a real number x with $|x - \pi| < \delta$ and $|\sin(x)| \geq \epsilon$.